Computer Networks

Lecture 1: Layering and Protocols

Demo Lecturer

Semester 2, 2026

What Is a Network?

- **Hosts** (end systems) run applications
- **Links** carry bits between nodes (copper, fibre, radio)
- **Switches and routers** forward data towards its destination

The hard part is not any single piece; it is making millions of independently built pieces work together.

Why Layers? A Postal Analogy

Sending a letter abroad: you write it, the post office sorts it, a plane carries it.

- You do not know which plane is used – and you should not need to
- Each step offers a **service** to the step above and uses the one below
- Any step can be replaced (a train instead of a plane) without changing the others

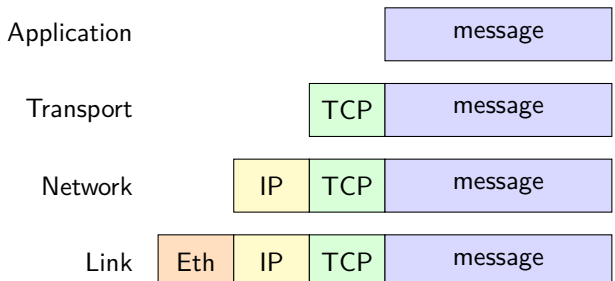
Layering = the same idea applied to network software.

The Five-Layer Model

Layer	Job	Example protocol	Unit of data
Application	what the user program says	HTTP, DNS	message
Transport	process-to-process delivery	TCP, UDP	segment
Network	host-to-host delivery across networks	IP	packet
Link	delivery over one hop	Ethernet, Wi-Fi	frame
Physical	bits on the medium	copper, fibre, radio	bit

Encapsulation

Each layer wraps the data from above with its own **header**.



Protocol, Service, Interface

Term	Meaning
Service	what a layer *offers* to the layer above
Interface	how the layer above asks for that service
Protocol	the rules two *peer* layers on different machines follow to talk

Changing a protocol inside one layer should not break the layers around it.

Summary and Next Time

- Layers hide detail: each offers a service upward and uses the service below
- Data is wrapped by one header per layer on the way down (encapsulation)
- Next lecture: the **physical layer** – how bits actually travel, and the limits on how fast